5) $f: \mathbb{R} \to \mathbb{R}, f(x,y) = \begin{cases} x^4 y \end{cases}$ ...

SEMINAR 9

COORDONATE POLARE

Fie $M(x,y)$

$x, y \to$ coord carteziene

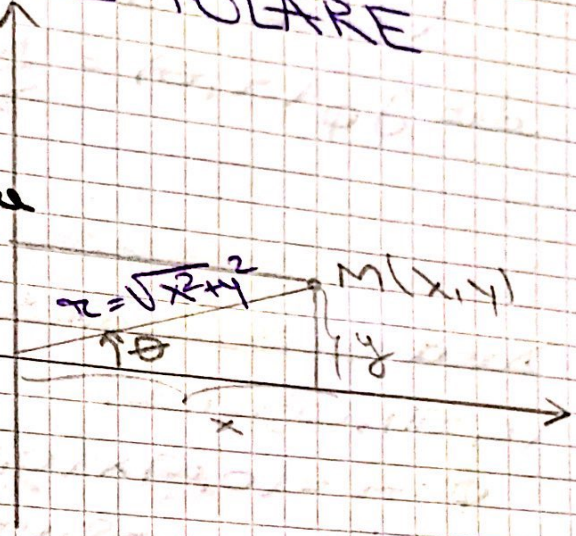
$OM = r = \sqrt{x^2+y^2}$

$\cos\theta = \frac{x}{r} \Rightarrow$

$\Rightarrow \boxed{x = r\cos\theta}$

$\sin\theta = \frac{y}{r} \Rightarrow$ (*)

$\Rightarrow \boxed{y = \sin\theta}$



- Dc $M = (0,0) \Rightarrow r = 0 \Rightarrow$ (*) adev $\forall \theta \in \mathbb{R}$
- Dc $M \neq (0,0)$, θ îl alegem într-un interval de lungime 2π, de obicei $[0, 2\pi)$ sau $[-\pi, \pi)$, atunci scrierea este unică

$\boxed{r, \theta \longrightarrow \text{coordonate polare pt } M}$

Considerând fct. $\Phi: [0,\infty) \times \mathbb{R} \to \mathbb{R}^2$,
$\Phi(r,\theta) = (r\cos\theta, r\sin\theta)$

(Scriem $\Phi: \begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$),

aplicația $(0,\infty) \times [0, 2\pi) \ni (r,\theta) \xrightarrow{\Phi} (r\cos\theta, r\sin\theta) \in \mathbb{R}^2 \setminus \{(0,0)\}$ este o bijecție

ϕ este diferențiabilă pe $[0,\infty)\times\mathbb{R}$ și

$$J_\phi(r,\theta)=\begin{pmatrix} \frac{\partial x}{\partial r}(r,\theta) & \frac{\partial x}{\partial \theta}(r,\theta) \\ \frac{\partial y}{\partial r}(r,\theta) & \frac{\partial y}{\partial \theta}(r,\theta) \end{pmatrix}=$$

$$=\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

$$\det J_\phi(r,\theta)=\frac{D(x,y)}{D(r,\theta)}=\begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$
$$=r\neq 0$$
$$\forall (r,\theta)\in(0,\infty)\times\mathbb{R}$$

Dacă restricționăm fct. ϕ la $(0,\infty)\times(0,2\pi)$, atunci:

$\phi: A=(0,\infty)\times(0,2\pi)\to\mathbb{R}^2\setminus\{(x,0);\ x\geq 0\}$

este difeomorfism (ie sch. de coordonate), deoarece:

1) ϕ bij

2) ϕ de clasă C^1

3) $\det J_\phi(r,\theta)\neq 0$, $\forall (r,\theta)\in A$.

$\boxed{\text{Ex. 1}}$

a) Arătați că funcția $f:\mathbb{R}^2\to\mathbb{R}^2$, $F(x,y)=(y-2x,\ y+2x)$ este difeomorfism de clasă C^1 folosind sch. de coordonate $\begin{cases} u=y-2x \\ v=y+2x \end{cases}$

b) Folosind această schimbare, transformați ecuația:

$$\frac{\partial^2 f}{\partial x^2}-4\frac{\partial^2 f}{\partial y^2}=0$$ și găsiți sol. generală.

SOL:

a) Obs că F este bijectivă, $F \in \mathcal{C}^1(\mathbb{R}^2)$

$$J_F(x,y) = \begin{pmatrix} -2 & 1 \\ 2 & 1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow \det J_F(x,y) = -4 \neq 0, \forall (x,y) \in \mathbb{R}^2 \Rightarrow$$

$\Rightarrow F$ ~~difeomorfism~~ difeomorfism

b) $\begin{cases} u = y - 2x \\ v = y + 2x \end{cases}$ (+)

$$\Rightarrow u + v = 2y \Rightarrow y = \frac{u+v}{2}$$

$$u - v = -4x \Rightarrow x = \frac{v-u}{2}$$

Caut $f(x,y) = g(u(x,y), v(x,y))$

- $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial x} =$

$$= -2 \frac{\partial g}{\partial u} + 2 \frac{\partial g}{\partial v}$$

- $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial y} =$

$$= \frac{\partial g}{\partial u} + \frac{\partial g}{\partial v}$$

- $\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(-2 \frac{\partial g}{\partial u} + 2 \frac{\partial g}{\partial v}\right) =$

$$= \frac{\partial}{\partial u}\left(-2 \frac{\partial g}{\partial u} + 2 \frac{\partial g}{\partial v}\right) \cdot \underbrace{\frac{\partial u}{\partial x}}_{-2} +$$

$$+ \frac{\partial}{\partial v}\left(-2 \frac{\partial g}{\partial u} + 2 \frac{\partial g}{\partial v}\right) \cdot \underbrace{\frac{\partial v}{\partial x}}_{2} =$$

$$= 4 \frac{\partial^2 g}{\partial u^2} - 4 \frac{\partial^2 g}{\partial u \partial v} - 4 \frac{\partial^2 g}{\partial v \partial u} + 4 \frac{\partial^2 g}{\partial v^2} =$$

$$= 4 \frac{\partial^2 g}{\partial u^2} - 8 \frac{\partial^2 g}{\partial u \partial v} + 4 \frac{\partial^2 g}{\partial v^2} \quad (1)$$

- $\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial g}{\partial u} + \frac{\partial g}{\partial v}\right) =$

$$= \frac{\partial}{\partial u}\left(\frac{\partial g}{\partial u} + \frac{\partial g}{\partial v}\right) \cdot \underbrace{\frac{\partial u}{\partial y}}_{1} + \frac{\partial}{\partial v}\left(\frac{\partial g}{\partial u} + \frac{\partial g}{\partial v}\right) \cdot \underbrace{\frac{\partial v}{\partial y}}_{1}$$

$$= \frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial u \partial v} + \frac{\partial^2 g}{\partial v \partial u} + \frac{\partial^2 g}{\partial v^2} =$$

$$= \frac{\partial^2 g}{\partial u^2} + 2 \cdot \frac{\partial^2 g}{\partial u \partial v} + \frac{\partial^2 g}{\partial v^2}. \quad (2)$$

"(1) − 4(2)" $\Rightarrow \frac{\partial^2 f}{\partial x^2} - 4 \cdot \frac{\partial^2 f}{\partial y^2} = 4 \cancel{\frac{\partial^2 g}{\partial u^2}} - 8 \frac{\partial^2 g}{\partial u \partial v} +$

$$+ 4 \cancel{\frac{\partial^2 g}{\partial v^2}} - 4 \cancel{\frac{\partial^2 g}{\partial u^2}} - 8 \frac{\partial^2 g}{\partial u \partial v} - 4 \cancel{\frac{\partial^2 g}{\partial v^2}} =$$

$$= -16 \frac{\partial^2 g}{\partial u \partial v}$$

Din ipoteză $MS = 0$ $\Big\} \Rightarrow$

$$\Rightarrow \frac{\partial^2 g}{\partial u \partial v} = 0.$$

$g(u,v) = ?$ a.î.

$$\Rightarrow \frac{\partial}{\partial u}\left(\frac{\partial g}{\partial v}\right) = 0 \Rightarrow \frac{\partial g}{\partial v}(u,v) = h(v) \Rightarrow$$

$\Rightarrow g(u,v) = H(v) + G(u)$ cu H, G de clasă C^2

$$\Rightarrow \boxed{f}(x,y) = g(u(x,y), v(x,y)) =$$

$$= H(v(x,y)) + G(u(x,y)) =$$

$$= \boxed{H(y+2x) + G(y-2x)}$$

Ex. 2

Fie f o fct. de clasă C^2 pe o mulț. deschisă din $\mathbb{R}^2 \setminus \{(0,0)\}$. Să se exprime laplacianul:

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$ în coord. polare.

Det. pct. curbei $x^2+xy+y^2=1$ care sunt cele mai depărtate de origine.

$\{(x,1)\in\mathbb{R}^2 \mid x\in[0,1]\}$

<u>Sol:</u>

Fie $f(x,y)=x^2+y^2=d^2((0,0),A(x,y))$

Fie $M=\{(x,y)\in\mathbb{R}^2 \mid x^2+xy+y^2=1\}$

$\sup\limits_{(x,y)\in M} f(x,y)=?$

Fie $g:\mathbb{R}^2\to\mathbb{R}$, $g(x,y)=x^2+xy+y^2-1$

$\operatorname{rang}\begin{pmatrix}\frac{\partial g}{\partial x} & \frac{\partial g}{\partial y}\end{pmatrix}=\operatorname{rang}(2x+y \quad 2y+x)=1$

$\forall (x,y)\in M$

• Fie $L(x,y)=f(x,y)+\lambda g(x,y)=$
$=x^2+y^2+\lambda(x^2+xy+y^2-1)$.

• Rez. sistemul $\begin{cases}\frac{\partial L}{\partial x}=0\\ \frac{\partial L}{\partial y}=0\\ g(x,y)=0\end{cases} \Rightarrow$

$\Rightarrow \begin{cases}2x+2\lambda x+\lambda y=0\\ 2y+2\lambda y+\lambda x=0\\ x^2+xy+y^2=1.\end{cases} \Rightarrow$

$\begin{cases}2x+\lambda(2x+y)=0\\ 2y+\lambda(2y+x)=0.\end{cases} \Big| \ominus \Rightarrow$

$\Rightarrow (x-y)(2+\lambda)=0 \begin{cases}\text{I } x=y\\ \text{II } \lambda=-2.\end{cases}$

I $\quad x=y \Rightarrow x^2+x^2+x^2=1 \Rightarrow x=\pm\frac{1}{\sqrt{3}}$

$\Rightarrow \left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\left(-\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}\right)$

$\frac{2}{\sqrt{3}}+\lambda\cdot\frac{2}{\sqrt{3}}+\lambda\cdot\frac{1}{\sqrt{3}}=0 \Rightarrow \boxed{\lambda=-\frac{2}{3}}$

(II) $\lambda = -2 \Rightarrow -2x - 2y = 0 \Rightarrow x = -y \Rightarrow$

$\Rightarrow \cancel{x^2} - \cancel{x^2} + x^2 = 1 \Rightarrow$

$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow$

$\Rightarrow (1,-1), (-1,1)$

- $d^2L(x_i, y_i) = \frac{\partial^2 L}{\partial x^2}(x_i, y_i)\, dx^2 +$

$+ 2\frac{\partial^2 L}{\partial x \partial y}(x_i, y_i)\, dx\, dy + \frac{\partial^2 L}{\partial y^2}(x_i, y_i)\, dy^2 =$

$= (2+2\lambda)\, dx^2 + 2\lambda\, dx\, dy + (2+2\lambda)\, dy^2$

- $d^2L\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = d^2L\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) \overset{\lambda = -\frac{2}{3}}{=}$

$= \frac{2}{3} dx^2 - \frac{4}{3} dx\, dy + \frac{2}{3} dy^2 > 0$

- $d^2L(1,-1) = d^2L(-1,1) \overset{\lambda = -2}{=} -2dx^2 - 4dx\, dy - 2dy^2 < 0$

$\Rightarrow \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$ - pct de MINIM pt. f pe A

$\Rightarrow (1,-1), (-1,1)$ - pct de MAXIM pt f pe A

Dist. maximă $= \sqrt{f(-1,1)} = \sqrt{2}$, adică pt. punctele $(1,-1), (-1,1)$ de pe curbă

Ex.4 Dem. că distanța de la pct. (x_0, y_0, z_0) la planul P de ecuație $P: ax + by + cz + d = 0$ (cu $a^2+b^2+c^2 \neq 0$) este:

$$dist((x_0, y_0, z_0), P) = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2+b^2+c^2}}$$

SOL: Fie $f(x,y,z) = d^2((x_0,y_0,z_0),(x,y,z)) = (x-x_0)^2+(y-y_0)^2+(z-z_0)^2$

$A = \{(x,y,z) \in \mathbb{R}^3 \mid ax+by+cz+d=0\}$, $a^2+b^2+c^2 \neq 0$

$\inf\limits_{A} f(x,y,z) = ?$

Fie $g : \mathbb{R}^3 \to \mathbb{R}$, $g(x,y,z) = ax+by+cz+d$.

$f, g \in \underbrace{C^2(\mathbb{R}^3)}_{\text{m.d.}}$

- $\text{rang}\begin{pmatrix} \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} & \frac{\partial g}{\partial z} \end{pmatrix} = \text{rang}(a\ b\ c) = 1$

- Fie $L(x,y,z) = f(x,y,z) + \lambda g(x,y,z) =$
$= (x-x_0)^2+(y-y_0)^2+(z-z_0)^2 + \lambda(ax+by+cz+d)$

- $\begin{cases} \frac{\partial L}{\partial x} = 0 \\ \frac{\partial L}{\partial y} = 0 \\ \frac{\partial L}{\partial z} = 0 \\ g(x,y,z) = 0 \end{cases} \Rightarrow \begin{cases} 2(x-x_0)+\lambda a = 0 \\ 2(y-y_0)+\lambda b = 0 \\ 2(z-z_0)+\lambda c = 0 \\ ax+by+cz+d = 0 \end{cases}$

$\Rightarrow \begin{cases} x = -\frac{\lambda a}{2} + x_0 \\ y = -\frac{\lambda b}{2} + y_0 \\ z = -\frac{\lambda c}{2} + z_0 \end{cases}$

$\Rightarrow ax_0 - \frac{\lambda a^2}{2} + by_0 - \frac{\lambda b^2}{2} + cz_0 - \frac{\lambda c^2}{2} + d = 0$

$\Rightarrow ax_0+by_0+cz_0+d - \frac{\lambda}{2}(a^2+b^2+c^2) = 0$

$\Rightarrow \lambda = \frac{2(ax_0+by_0+cz_0+d)}{a^2+b^2+c^2} \Rightarrow$

$$\Rightarrow x' = -\frac{a\lambda}{2} + x_0 =$$

$$= x_0 - a \cdot \frac{(ax_0+by_0+cz_0+d)}{a^2+b^2+c^2}$$

$$y' = y_0 - \frac{b(ax_0+by_0+cz_0+d)}{a^2+b^2+c^2}$$

$$z' = z_0 - \frac{c(ax_0+by_0+cz_0+d)}{a^2+b^2+c^2}$$

$\Rightarrow (x', y', z')$ - pct. critic

$$d^2L(x', y', z') = 2dx^2 + 2dy^2 + 2dz^2 + 0 \ldots = 2(dx^2+dy^2+dz^2) > 0$$

$\Rightarrow (x', y', z')$ pct. critic pt. f pe A

$$\text{dist}((x_0, y_0, z_0), P) = \sqrt{f(x', y', z')} = \sqrt{(x'-x_0)^2 + (y'-y_0)^2 + (z'-z_0)^2} =$$

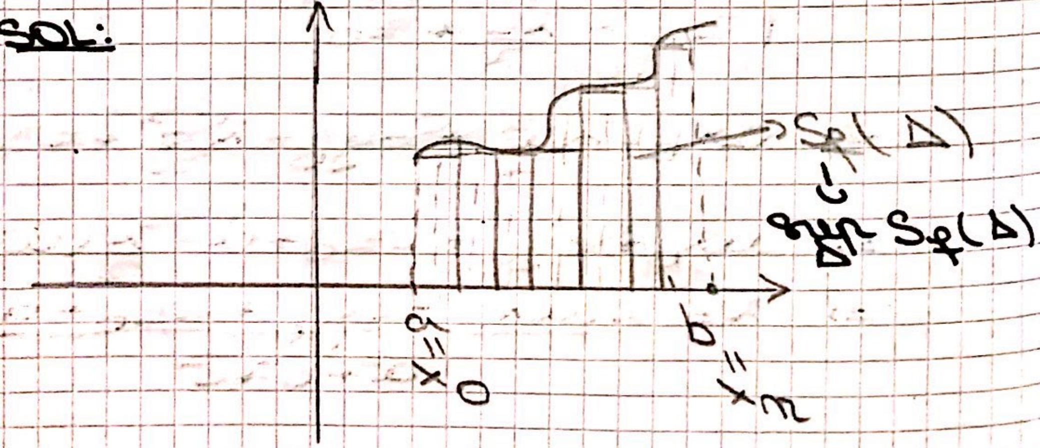
$$= \frac{|ax_0+by_0+cz_0+d|}{\sqrt{a^2+b^2+c^2}}.$$

$\square$

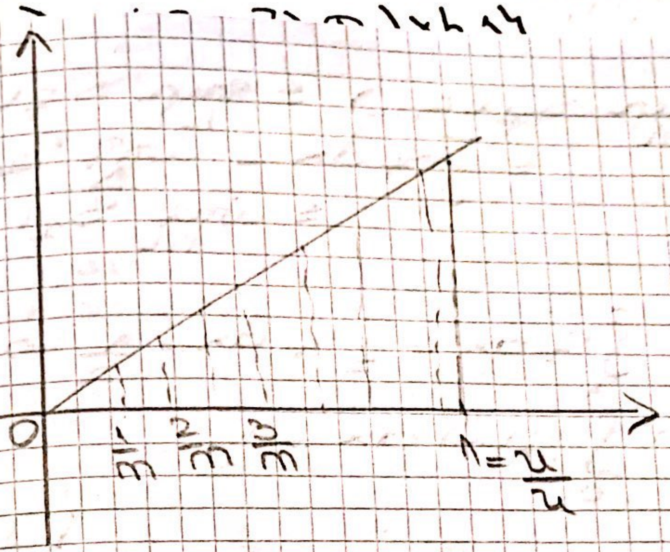
Ex.5

Dem. că funcția $f : [0,1] \to \mathbb{R}$, $f(x) = x$ este int (R) și calculați $\int_0^1 x\,dx$ folosind suma Darboux.

SOL:



4) $D = [0,1] \times \{1\} = \{(x,1) \in \mathbb{R}^2 \mid x \in [0,1]\} \subseteq \mathbb{R}^2$



Trebuie să arătăm că:

$$\underline{\int} f\,dx = \overline{\int} f\,dx.$$

$$x_i - x_{i-1} = \frac{1}{n}$$

Fie $\Delta_n = \left(0 < \frac{1}{n} < \frac{2}{n} < \ldots < \frac{n-1}{n} < \frac{n}{n} = 1\right) \in \mathcal{D}([0,1])$

$$m_i = \inf_{x\in[x_{i-1},x_i]} f(x) \underset{f\,\text{cont}}{=} \min_{[x_{i-1},x_i]} f(x) = f(x_{i-1})$$

$$M_i = \sup_{x\in[x_{i-1},x_i]} f(x) \underset{\substack{\text{cresc}\\ [0,1]\,\text{compact}}}{=} \max_{[x_{i-1},x_i]} f(x) = f(x_i)$$

$\Rightarrow m_i = x_{i-1},\ M_i = x_i.$

$$s_f(\Delta_n) = \sum_{i=1}^{n} (x_i - x_{i-1}) \cdot m_i = \sum_{i=1}^{n} \frac{1}{n} \cdot \frac{i-1}{n} = \frac{1}{n^2} \cdot \sum_{i=1}^{n} (i-1) = \frac{1}{n^{\cancel{2}}} \cdot \frac{\cancel{n}(n-1)}{2} = \frac{n-1}{2n}.$$

$$S_f(\Delta_n) = \sum_{i=1}^{n} (x_i - x_{i-1}) \cdot M_i = \sum_{i=1}^{n} \frac{1}{n} \cdot \frac{i}{n} = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}.$$

$$\frac{1}{2} = \sup_{n} s_f(\Delta_n) \leq \sup_{\Delta} s_f(\Delta_{ech}) = \underline{\int} f\,dx \leq \overline{\int} f\,dx = \inf_{\Delta} S_f(\Delta_{ech}) \leq \inf_{n} S_f(\Delta_n) = \frac{1}{2}$$

$$\Rightarrow \overline{\int} f\,dx = \underline{\int} f\,dx = \frac{1}{2} \Rightarrow$$

$$\Rightarrow \int_0^1 f(x)\,dx = \frac{1}{2}.$$

$\square$

Ex. 6.

Dem. folosind sume Darboux (def/crit), că $f: [0, \frac{\pi}{2}] \to \mathbb{R}$, $f(x) = \sin x$ este int (R) și calculați $\int_0^{\frac{\pi}{2}} f(x)\,dx$.

<u>Sol</u>:

$\boxed{T}$ $f: [a,b] \to \mathbb{R}$

f int (R) $\Leftrightarrow$ f mărg & $\forall \varepsilon > 0$, $\exists \delta_\varepsilon > 0$ a.î. $S_f(\Delta) - s_f(\Delta) < \varepsilon$, $\forall \Delta \in \mathcal{D}([a,b])$, $\|\Delta\| < \delta_\varepsilon$.

<u>Convexitate</u>:

Evid, f mărginită, $f(x) \in [0,1]$, $\forall x \in [0, \frac{\pi}{2}]$. Fie $\varepsilon > 0$ fixat.

$\delta_\varepsilon = ?$ a.î. $S_f(\Delta) - s_f(\Delta) < \varepsilon$, $\forall \Delta \in \mathcal{D}$, $\|\Delta\| < \delta_\varepsilon$

Fie $\Delta = (0 < x_0 < x_1 < \ldots < x_n = \frac{\pi}{2})$.

$m_i = \inf\limits_{x \in [x_{i-1}, x_i]} f(x) \overset{f \text{ cont}}{\underset{f \nearrow}{=\!=\!=}} \min f(x) = f(x_{i-1})$

$M_i = \sup\limits_{x \in [x_{i-1}, x_i]} f(x) \underset{\text{compact}}{=\!=\!=} \max f(x) = f(x_i)$

Deci $S_f(\Delta) - s_f(\Delta) =$

$= \sum\limits_{i=1}^{n} (x_i - x_{i-1}) \cdot M_i -$

$- \sum\limits_{i=1}^{n} (x_i - x_{i-1}) \cdot m_i =$

$= \sum\limits_{i=1}^{n} (x_i - x_{i-1})(\underbrace{M_i - m_i}_{\sin x_i - \sin x_{i-1}})$

$\leq \|\Delta\|$

$= \|\Delta\| \cdot \sum\limits_{i=1}^{n} (\sin x_i - \sin x_{i-1}) =$

$= \|\Delta\| (\sin \frac{\pi}{2} - \underbrace{\sin 0}_{0}) = \|\Delta\| \cdot \underbrace{\sin \frac{\pi}{2}}_{1}$

$= \|\Delta\|$ (1)

Aleg $\delta_\varepsilon = \frac{\varepsilon}{2} \Rightarrow f$ int(R).

- $\int_0^{\frac{\pi}{2}} f(x)\,dx = ?$

Alegem șirul un șir de diviziuni $(\Delta_n)_n \subset \mathcal{D}([0, \frac{\pi}{2}])$ cu $\|\Delta_n\| \xrightarrow{n} 0$.

astfel $S_f(\Delta_n) \xrightarrow{n} \int_0^{\frac{\pi}{2}} f(x)\,dx$.

Fie $\Delta_n = (0 < \frac{\pi}{2n} < \frac{2\pi}{2n} < \ldots < \frac{(n-1)\pi}{2n} < \frac{n\pi}{2n} = \frac{\pi}{2}) \in \mathcal{D}([0, \frac{\pi}{2}])$, $\forall\, n \in \mathbb{N}^*$

$\|\Delta_n\| = \frac{\pi}{2n} \xrightarrow{n \to \infty} 0$.

$$S_f(\Delta_n) = \sum_{i=1}^{n} \underbrace{(x_i - x_{i-1})}_{\frac{\pi}{2n}} \cdot \underbrace{M_i}_{= f(x_i) = \sin x_i = \sin\frac{i\pi}{2n}} =$$

$$= \frac{\pi}{2n} \cdot \sum_{i=1}^{n} \sin\frac{i\pi}{2n} =$$

$$= \frac{\pi}{2n}\left(\sin\frac{\pi}{2n} + \sin\frac{2\pi}{2n} + \dots + \sin\frac{n\pi}{2n}\right) \circledast$$

folosim formula:

$$\boxed{\begin{gathered}\sin x + \sin(x+k) + \sin(x+2k) + \\ \dots + \sin(x+nk) = \\ = \frac{\sin\left(\frac{n+1}{2}k\right)\cancel{\cos}\left(x+\frac{n}{2}k\right)}{\sin\frac{k}{2}}\end{gathered}}$$

La noi, $x = \frac{\pi}{2n}$, $k = \frac{\pi}{2n}$

$$\circledast = \frac{\pi}{2n} \cdot \frac{\sin\left(\frac{\cancel{n}}{2}\cdot\frac{\pi}{2\cancel{n}}\right) \cdot \cancel{\cos}\,\sin\left(\frac{\pi}{2n} + \frac{n-1}{2}\right.}{\sin\frac{\pi}{4n}}$$

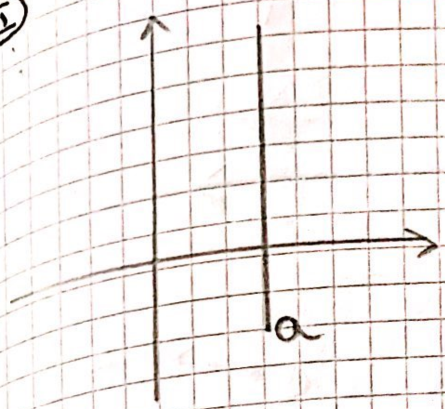
$$= \frac{\pi}{2n} \cdot \frac{\sin\left(\frac{\pi}{4}\right) \cdot \sin\left(\frac{(n+1)\pi}{4n}\right)}{\sin\frac{\pi}{4n}} =$$

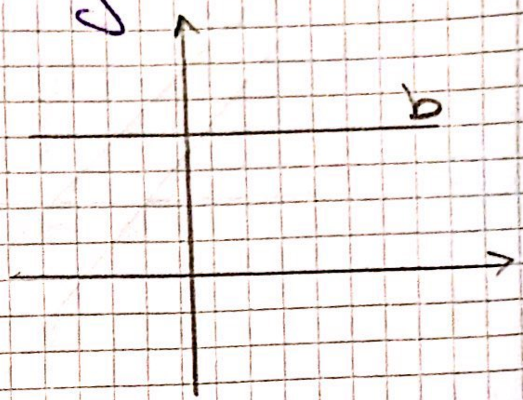
$$= \underbrace{\frac{\pi}{2n}}_{\to 0} \cdot \frac{\frac{\sqrt{2}}{2} \cdot \sin\frac{(n+1)\pi}{4n}}{\underbrace{\frac{\sin\frac{\pi}{4n}}{\frac{\pi}{4n}}}_{\to 1} \cdot \cancel{\frac{\pi}{4n}}} \qquad \sin\frac{(n+1)\pi}{4n} \to \frac{\sqrt{2}}{2}, \quad \sin\frac{\pi}{4n} \to 0$$

$$\xrightarrow{n \to \infty} \left(\frac{\sqrt{2}}{2}\right)^2 \cdot 2 = 1$$

RECAP. LICEU (REP. GRAFICE)

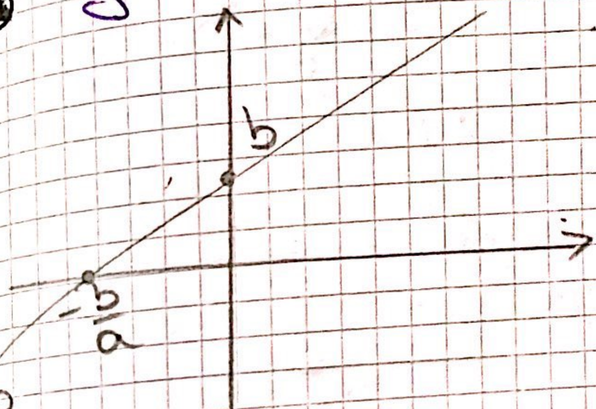
(I) $x = a$ $y = b$



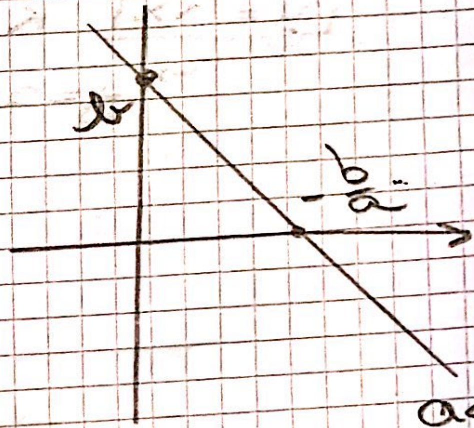


$Oy: x = 0$ $Ox: y = 0.$

(II) $y = ax + b \longrightarrow$ dreaptă



$a > 0$



$a < 0.$

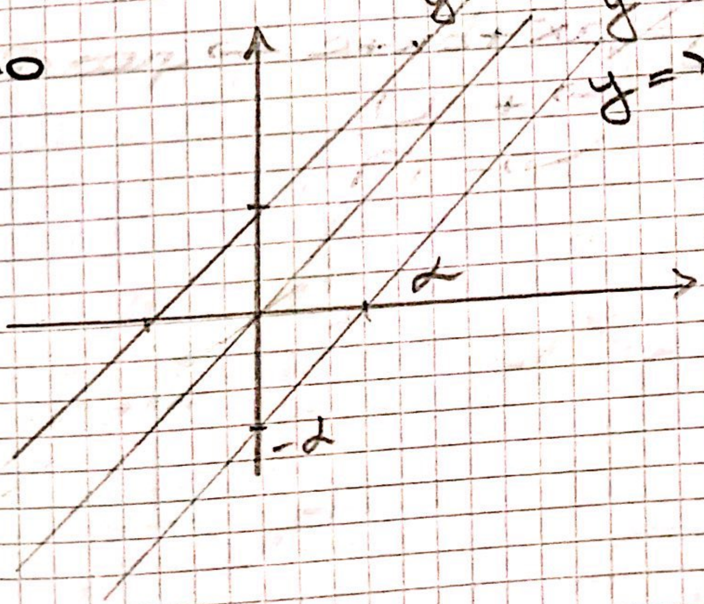
Cazuri particulare: $x+\alpha$

$y = x + \alpha$

$y = x$

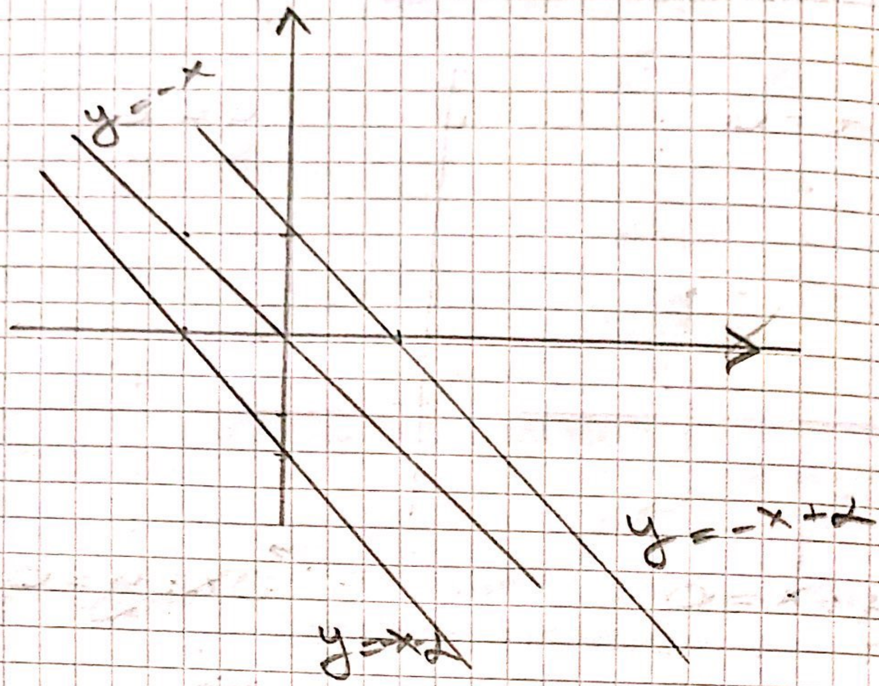
$y = x - \alpha$

$\alpha > 0$



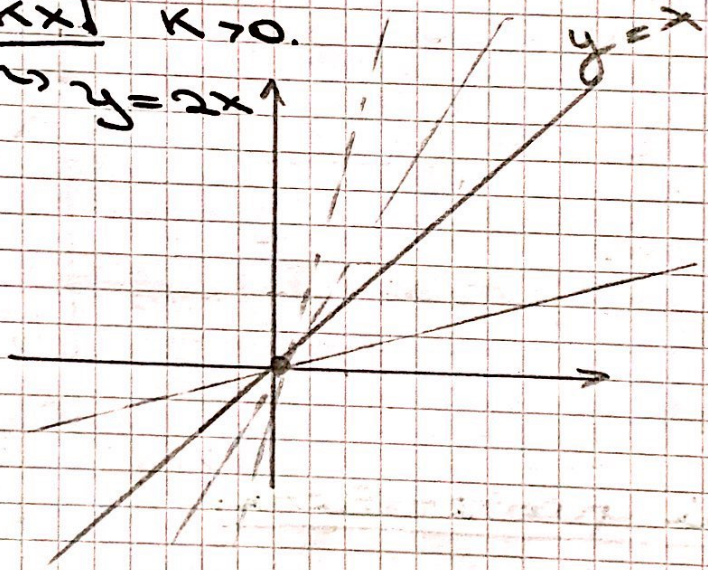
Scanned with CamScanner

$\alpha < 0$



$\boxed{y = kx}$ $k > 0.$

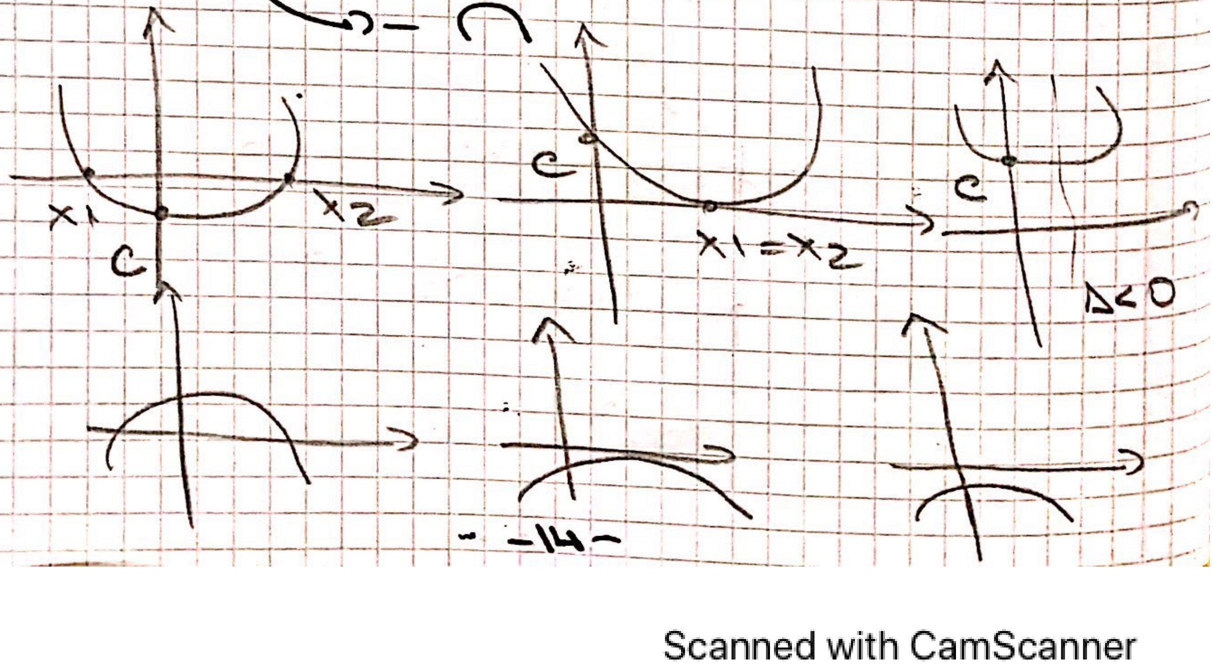
$\rightarrow y = 2x$



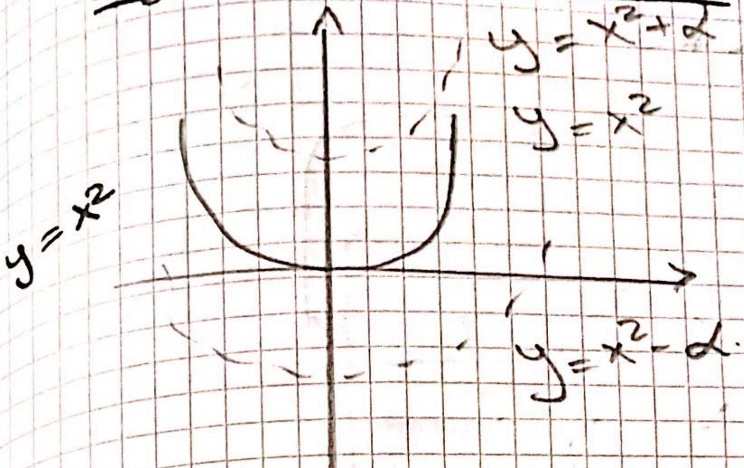
Ⅲ $y = ax^2 + bx + c \rightarrow$ parabola

$\rightarrow + \ \cup$

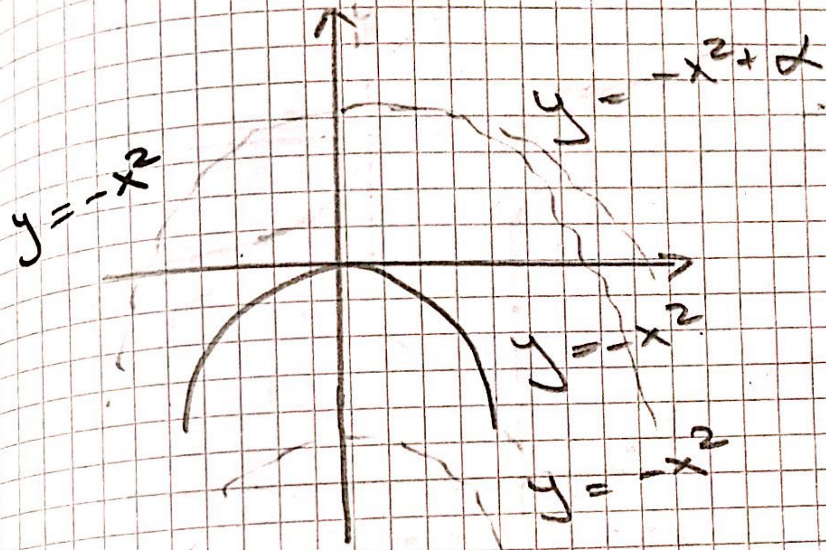
$\rightarrow - \ \cap$



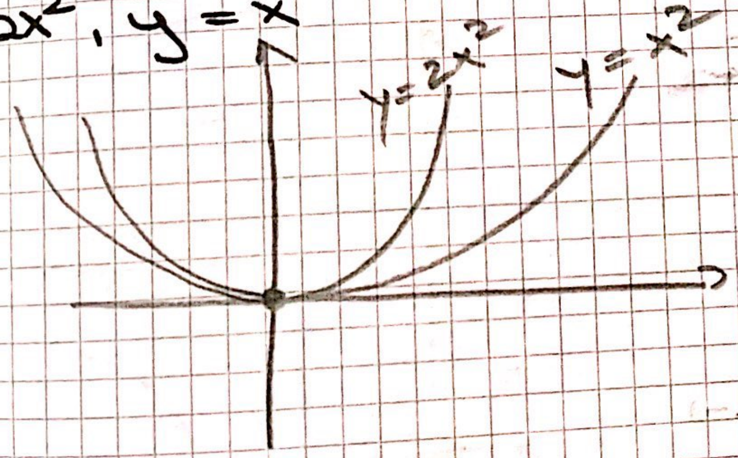
Cazuri particulare:



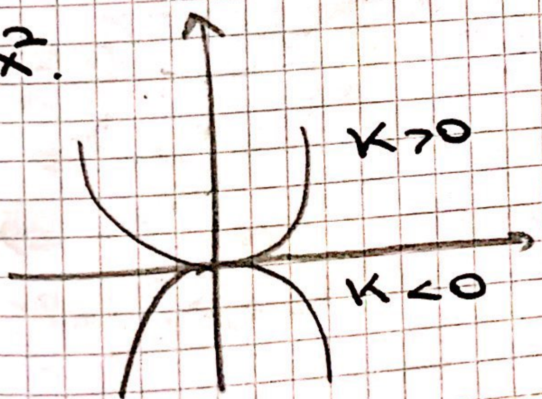
$\alpha > 0$



$y = 2x^2$, $y = x^2$

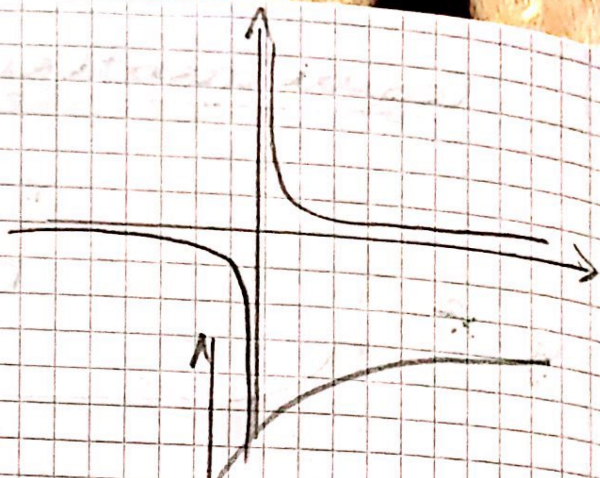


$y = Kx^2$.

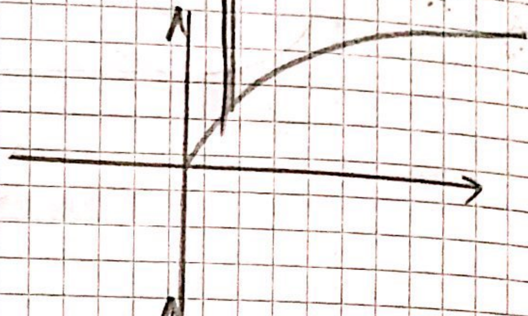


„Dvioare"

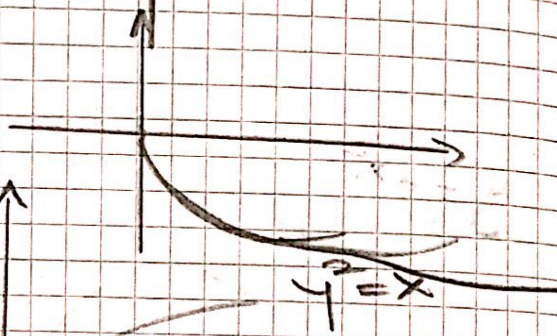
$y=\frac{1}{x},\ x\neq 0$



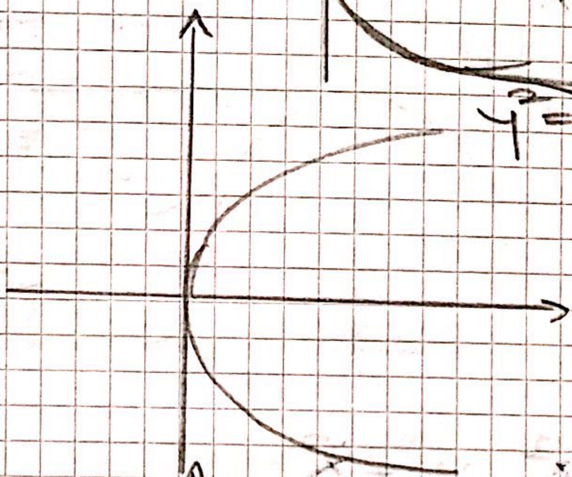
$y=\sqrt{x}$



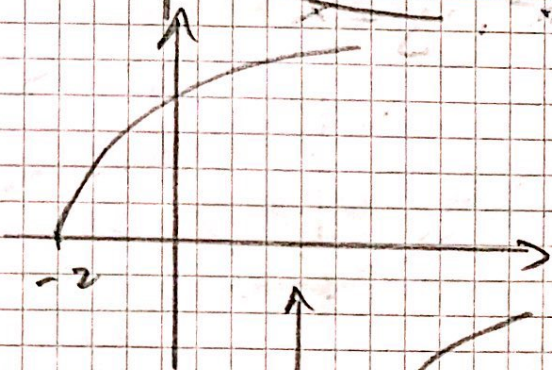
$y=-\sqrt{x}$



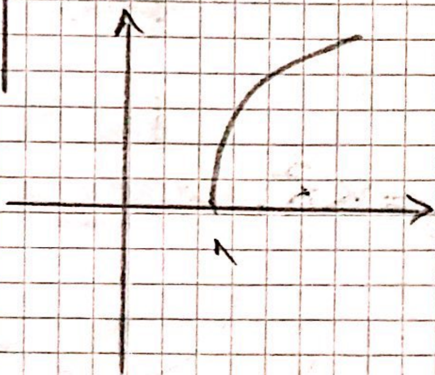
$y^2=x$



$y=\sqrt{x+2}$

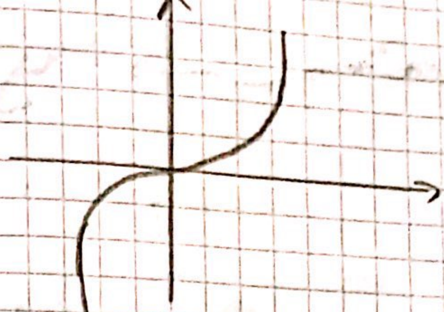


$y=3\sqrt{x-1}$

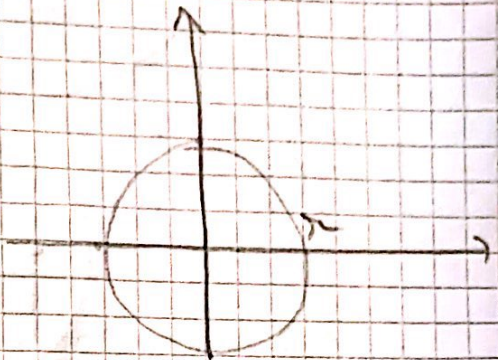


Scanned with CamScanner

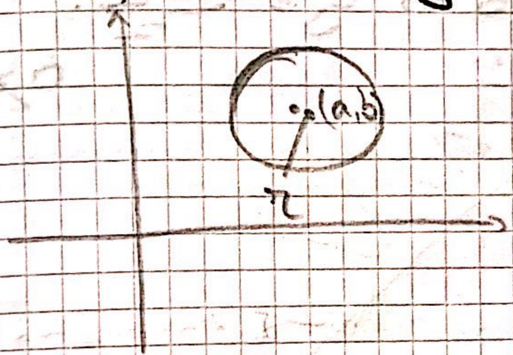
$y = x^3$



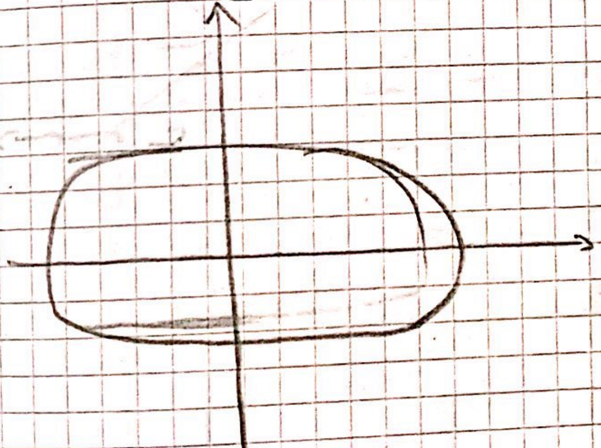
cercle $\mathcal{C}(O, r)$ $\quad x^2+y^2=r^2$



$\mathcal{C}((a,b), r)$ $\quad (x-a)^2+(y-b)^2=r^2$



Elipse: $\quad \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$



$$\frac{(x-x_0)^2}{a^2}+\frac{(y-y_0)^2}{b^2}=1.$$

Exp: $x^2+y^2=2x-4y$

$(x-1)^2+(y+2)^2=(\sqrt{5})^2$

$\Rightarrow \mathcal{C}((1,-2), \sqrt{5})$

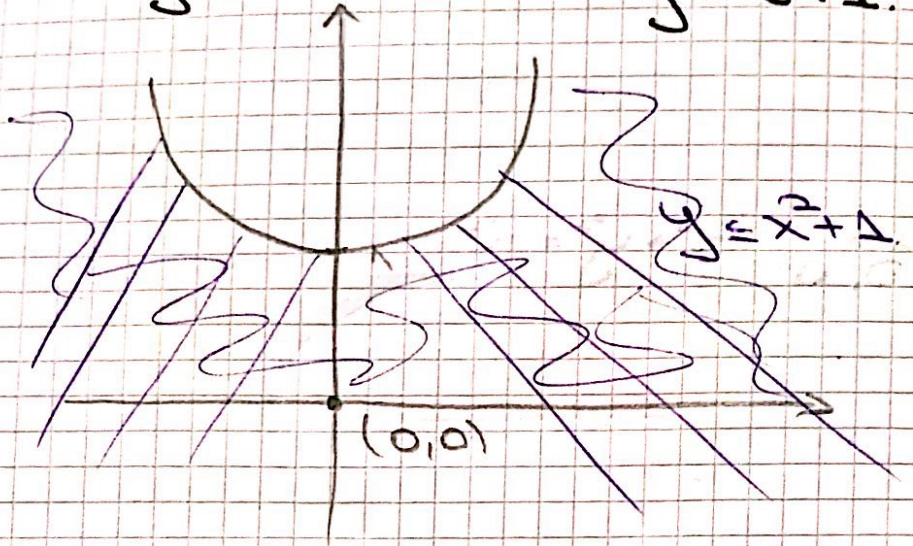
$x^2+4y^2=16$

$\frac{x^2}{16}+\frac{y^2}{4}=1$

$\left(\frac{x}{4}\right)^2+\left(\frac{y}{2}\right)^2=1$

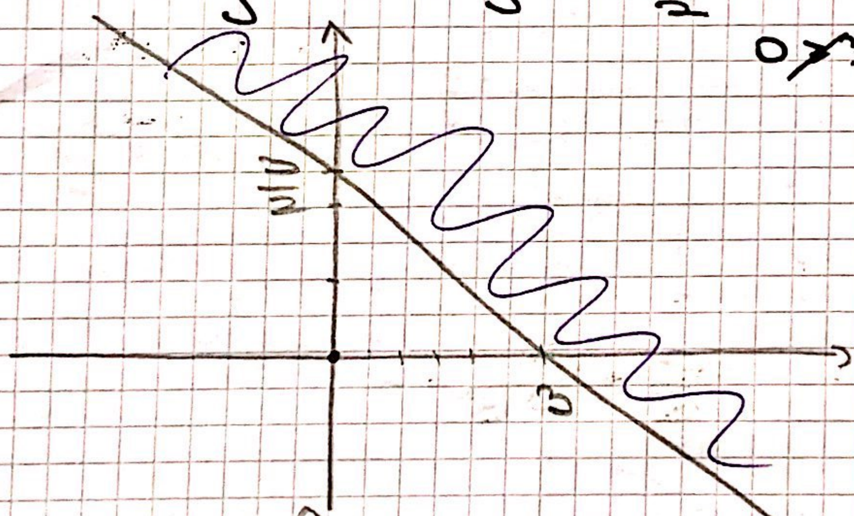
$\Rightarrow$ Elipse

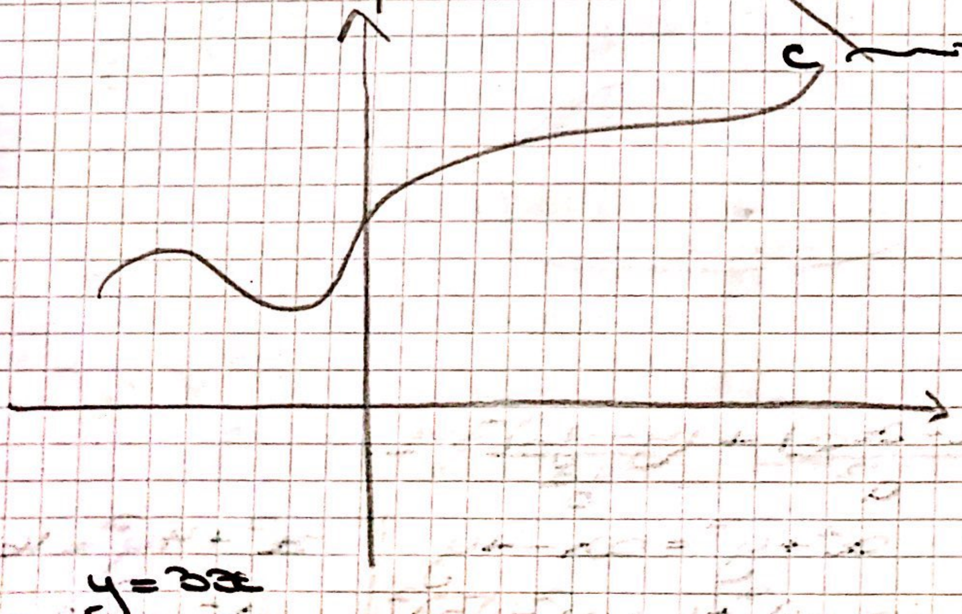
$0 \le y \le x^2+1 \rightsquigarrow y = x^2+1.$



$x + 2y \geq 3 \Rightarrow y \geq \frac{3-x}{2}.$

$0 \ngeq \frac{3}{2}$

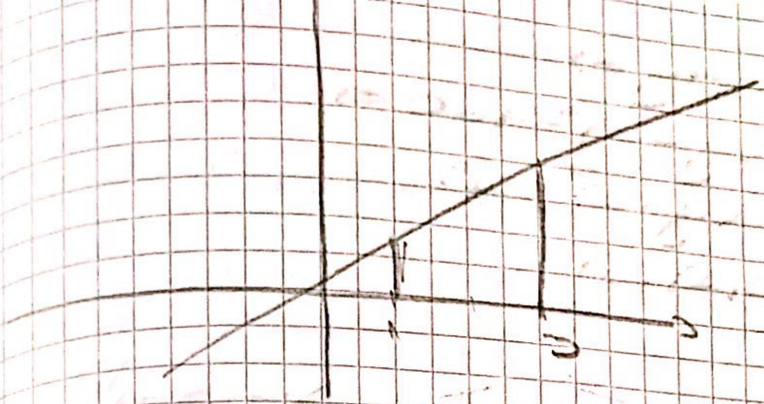




$C \rightsquigarrow$ Cost = ?

$y = 3x$

$x \in [1,3]$

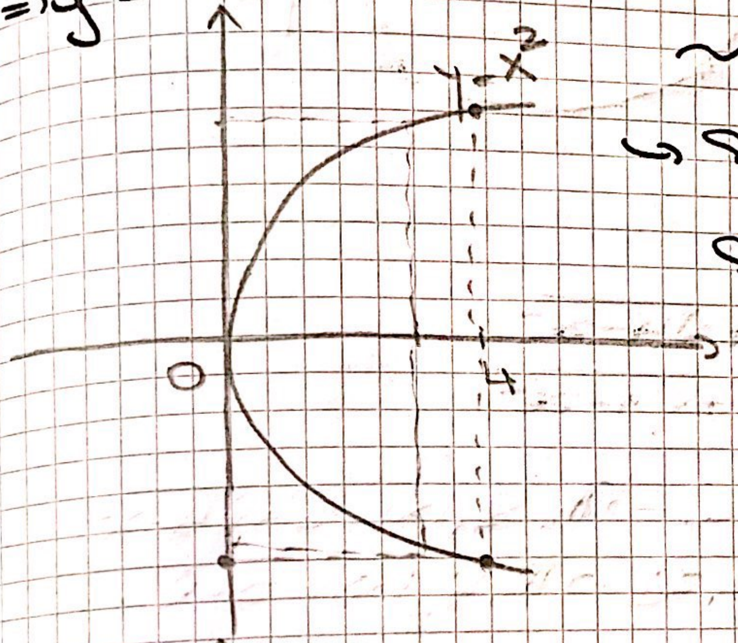


$f : [1,3] \rightarrow \mathbb{R}^2$

$f(x) = 3x.$

$x = y^2 , x \in [0,4]$

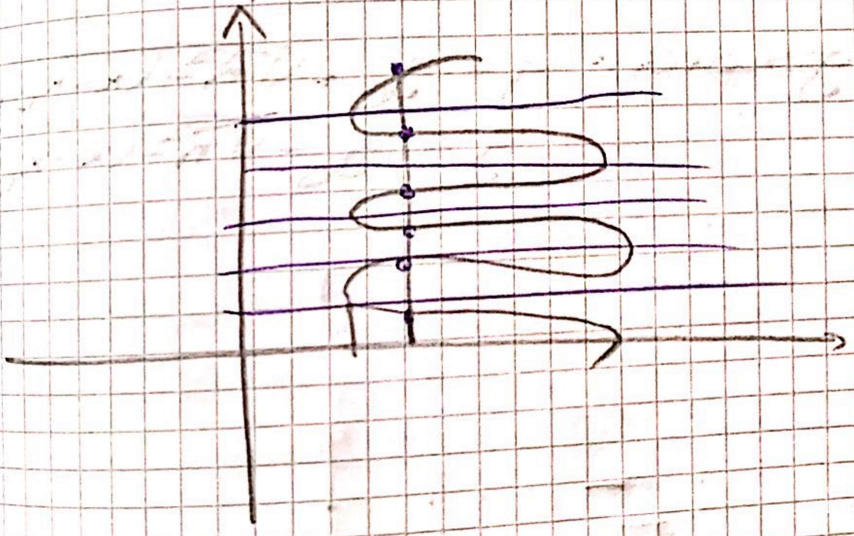
$\Rightarrow y = \pm \sqrt{x}.$



$\rightsquigarrow$ dlue e G_f

$\hookrightarrow f_1 : [0,4] \rightarrow \mathbb{R},$
$f_1(x) = \sqrt{x}$

$f_2 : [0,4] \rightarrow \mathbb{R}$
$f_2(x) = -\sqrt{x}.$

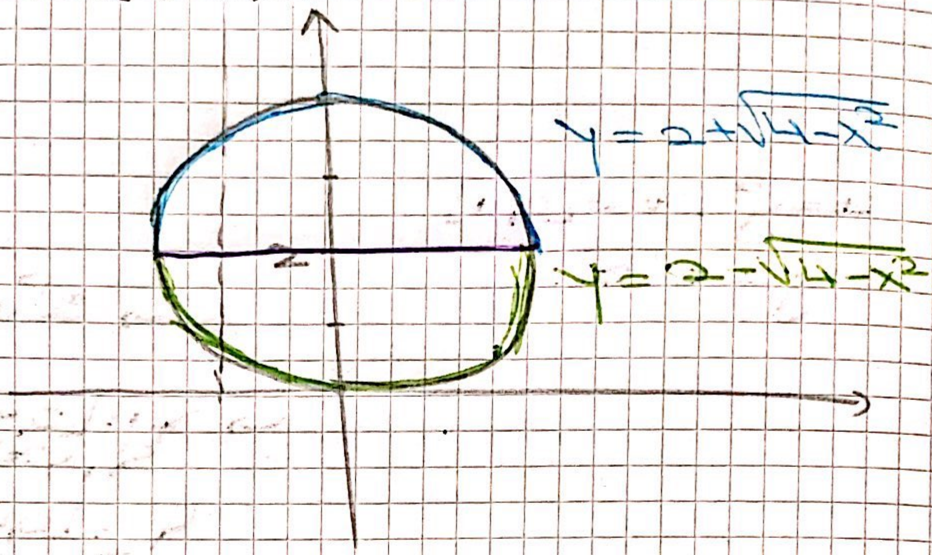


$x^2+y^2=4y \Rightarrow$

$\Rightarrow x^2+(y-2)^2-4=0 \Rightarrow$

$\Rightarrow x^2+(y-2)^2=2^2.$

$\mathcal{C}((0,2),2)$



$y-2=\pm\sqrt{4-x^2}$

$y=2\pm\sqrt{4-x^2}$

$f_1:[-2,2]\to\mathbb{R},\ f_1(x)=2+\sqrt{4-x^2}$

$f_2:[-2,2]\to\mathbb{R},\ f_2(x)=2-\sqrt{4-x^2}$

In general: $y=y_0+\sqrt{R^2-(x-x_0)^2}$

$y=y_0-\sqrt{R^2-(x-x_0)^2}$